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REVIEW

High-Throughput Methods for Variant Functional Assessment in Cardiac Disease

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ABSTRACT: In vitro functional modeling of genetic variants has revolutionized our understanding of which variants can cause cardiac disorders, providing insights into their molecular underpinnings. This review provides an overview of high-throughput methods used for the functional assessment of variants implicated in inherited cardiac diseases. Advances in gene-editing technology now enable the efficient generation of cells expressing individual genetic variants or libraries of variants for robust functional studies. We discuss innovative assays that can evaluate dozens or hundreds of variants sequentially. For example, the electrophysiological properties of numerous cardiac ion channel variants in genes linked to inherited arrhythmias can be characterized using automated patch clamping. The mechanical properties of cardiomyocytes expressing candidate cardiomyopathy variants can be assessed using techniques such as atomic force microscopy, traction force microscopy, and impedance-based methods. Multiplexed assays of variant effect are an emerging family of techniques that use gene-specific or general assays, combined with next-generation sequencing, to characterize hundreds or thousands of pooled genetic variants. We examine the key advantages and limitations of each method and outline future goals for the field. Innovative in vitro studies of cardiac genetic variants will enhance our understanding of variant-disease relationships and improve diagnosis, screening, and treatment options for these disorders.

Key Words: cardiomyopathies ■ CRISPR-Cas systems ■ high-throughput screening assays ■ induced pluripotent stem cells ■ mechanics ■ patch-clamp techniques

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Cardiovascular diseases (CVDs) are responsible for approximately one-third of global deaths.[1] This group of disorders involves the heart, brain, and blood vessels, contributing to the burden of heart failure, sudden cardiac death, and stroke. While behavioral factors such as tobacco use, physical inactivity, and environmental factors such as pollution modify the risk of CVDs, many of these diseases have a strong genetic component. CVDs such as hypertrophic cardiomyopathy (HCM), dilated cardiomyopathy, and inherited arrhythmia syndromes have been attributed to disease-causing variants, primarily in genes encoding sarcomeric proteins and voltage-gated ion channels. The American College of Medical Genetics and Genomics variant classification framework categorizes variants as pathogenic, likely pathogenic, variant of uncertain significance (VUS), likely benign, or benign.[2] The discovery of a pathogenic variant in specific genes can often be clinically actionable, informing diagnosis, medical management, and family screening. Current guidelines recommend the reporting of incidentally detected pathogenic variants in 81 clinically actionable genes, including 41 genes associated with CVD.[3] In recent years, the cost of clinical genetic testing has decreased dramatically. However, many genetic testing results return VUS, which are not clinically actionable; this is known as the VUS problem.[2] High-throughput assays can provide functional evidence for the reclassification of VUS and improve the diagnosis and treatment of cardiac diseases. In addition, functional assays play a critical role in establishing and validating gene–disease relationships. The Clinical Genome Resource is spearheading systematic efforts to both confirm novel associations and reassess genes previously implicated in Mendelian cardiac disorders.[4,5] While functional data are

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Nonstandard Abbreviations and Acronyms

AFM	atomic force microscopy
CRISPR-Cas9	clustered regularly interspaced short palindromic repeats/clustered regularly interspaced short palindromic repeat–associated 9
CVD	cardiovascular disease
EHT	engineered heart tissue
GFP	green fluorescent protein
gRNA	guide RNA
HCM	hypertrophic cardiomyopathy
hiPSC-CM	human induced pluripotent stem cell–derived cardiomyocyte
MAVE	multiplexed assay of variant effect
nCas9	Cas9 nickase
TUNEL	terminal deoxynucleotidyl transferase dUTP nick end labeling
VUS	variant of uncertain significance
WT	wild type
β-MHC	β-myosin heavy chain

not used in isolation, they can serve as critical evidence in supporting these relationships.

We divide high-throughput variant studies into 2 main types: arrayed assays and pooled assays (Figure 1; Table 1). Arrayed assays represent a more traditional approach, where a series of variants are tested sequentially or in multiwell plates, where each well contains cells expressing a single variant.[37] Platforms such as automated patch-clamp devices, modified traction force microscopy instruments, and plate readers can accommodate experiments using 96- or 384-well plates; certain plate readers support up to 1536 wells. Although the throughput of these assays can be high, it is typically limited by the number of generated variants, the number of available wells, and the need for replicates. Quantitative data can be collected for various cellular properties, including ion influx, contractile force, or patch-clamp measurements, enabling nuanced mechanistic studies with direct functional comparisons between variants.[14,21]

Pooled assays, often referred to as multiplexed assays of variant effect (MAVEs), enable the simultaneous study of variant effects on molecular phenotypes such as protein function, gene expression, or mRNA splicing.[38] In an MAVE experiment, comprehensive libraries of variants are expressed in cells, subjected to multiplexed assays of properties such as fluorescence intensity or cell survival, and then deep sequenced to quantify variant-specific functional effects. Multiple MAVE assays for a single gene can provide key information about variant properties or identify mechanisms of variant pathogenicity. Deep mutational scanning is a subtype of MAVE focused on protein-coding variants. Standard methods include survival assays or fluorescence-based assays to separate cells based on their variant effects.[7] Pooled assays measure variant effects in identical conditions, reducing environmental variation. While pooled assays provide high-throughput data sets, array-based assays currently offer a wider range of functional assay types. Notably, certain assays are not yet compatible with pooled assays with current technologies. The choice between the 2 approaches depends on specific experimental goals.

Regardless of the assay type, a central challenge is efficiently generating cells that express genetic variants. In the first section, we review current strategies for introducing single or pools of genetic variants into cells. We then discuss a variety of arrayed assays used to study cardiac phenotypes. Last, we discuss pooled MAVE assays and their emerging role in characterizing hundreds to thousands of variants in cardiac disease genes.

GENERATION OF VARIANT-EXPRESSING CELLS

A key aspect of high-throughput variant assays is the efficient generation of cells expressing genetic variants. While transient transfections are efficient and fast, this approach is limited due to the inability to rigorously standardize expression levels.[39] Instead, creating cell lines that stably and consistently express different variants at the endogenous genomic locus or a landing pad site can be a more reliable approach. Figure 2 depicts commonly used methods to single-variant–expressing cells or pools of variant-expressing cells.

Classic CRISPR-Cas9 (clustered regularly interspaced short palindromic repeats/clustered regularly interspaced short palindromic repeat–associated 9) gene editing involves introducing a guide RNA (gRNA) with Cas9 that causes a double-strand break at a specific locus. If a repair template is provided, the homology-directed repair pathway repairs the DNA lesion and introduces a variant of interest.[40] Saturation genome editing uses CRISPR-Cas9 to generate a double-strand break in a gene of interest, followed by homology-directed repair from a pool of single-stranded oligodeoxynucleotide sequences containing a library of variants. This approach has been used in haploid cells, typically the HAP1 cell line, to generate pools of cells expressing all possible variants at a locus.[41] Prime editing and base editing are 2 variants of CRISPR-Cas9 that can lead to enhanced genome editing efficiency and versatility.[42] Prime editing uses an nCas9 (Cas9 nickase) enzyme to make a single-stranded nick at a site directed by the prime editing gRNA. A reverse transcriptase enzyme fused to nCas9 then uses the prime editing gRNA as a template for variant synthesis.[43] This approach can also be used to introduce many genetic

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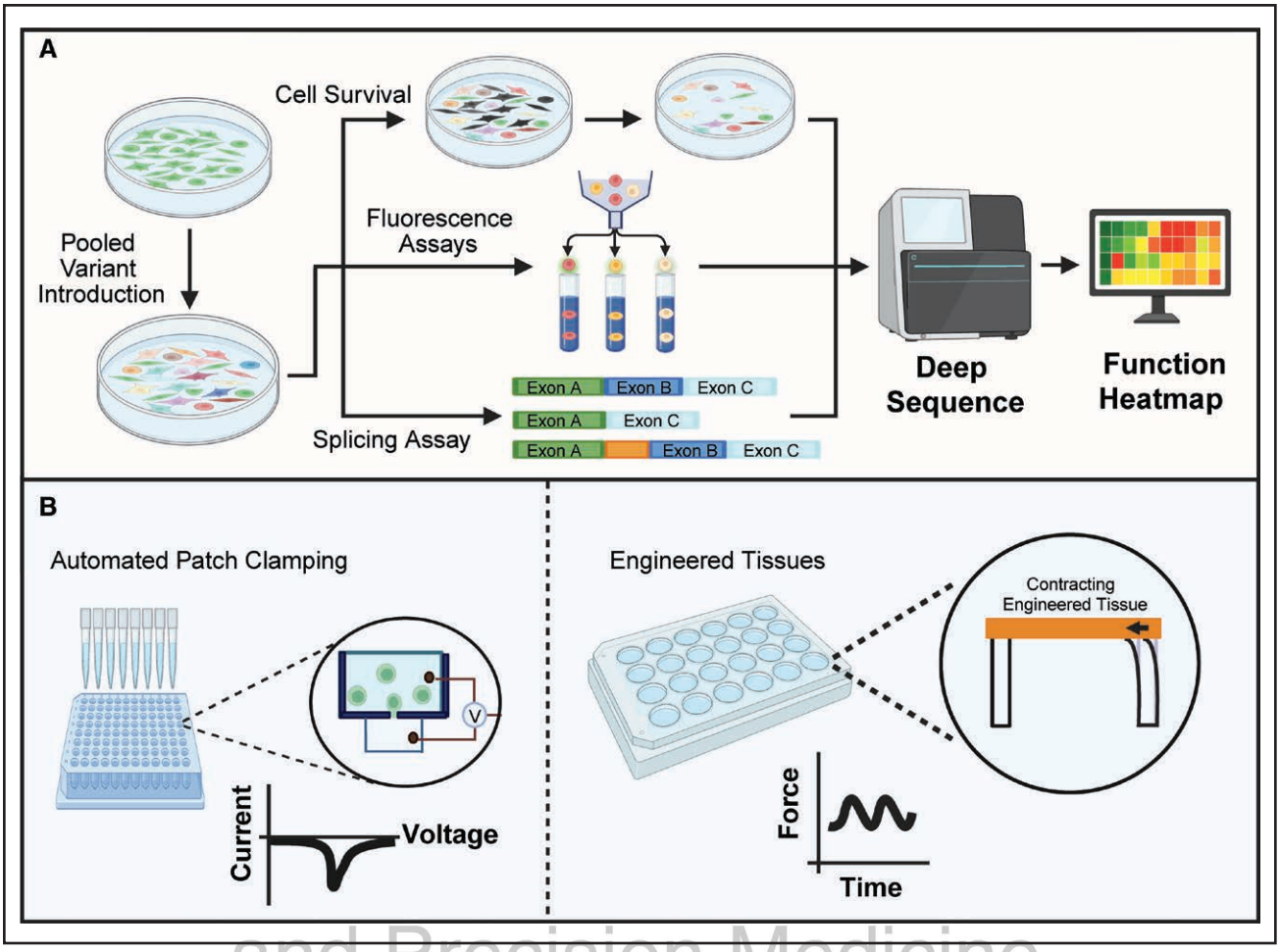

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Figure 1. Overview of pooled and arrayed cardiac functional assays.
A, Pooled assays. A library of variants is transfected into pooled cells and subjected to multiplexed assays that determine function scores for each variant, which are displayed on a heatmap. These assays include cell survival assays, fluorescence assays, and splicing assays. **B**, Arrayed assays. In arrayed assays, for example, automated patch clamping and engineered heart tissues, variants are serially measured in separate wells.

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variants simultaneously.[44,45] In recent work, a library of prime editing gRNAs was delivered to cells, and prime editing was used to generate thousands of variants in the epithelial growth factor receptor gene, which is implicated in cardiac remodeling and atrial fibrillation.[44,45] Base editing uses a catalytically dead Cas9 fused to a deaminase to chemically alter bases. Although no single base editing system can generate every possible modification, available tools can generate all 4 purine-to-purine and pyrimidine-to-pyrimidine transition modifications (C→T, A→G, G→A, and T→C).[46,47] Prime and base editing methods can reduce off- and on-target genomic changes by eliminating the need for double-strand breaks. An important limitation for all families of CRISPR gene-editing methods is the generation of undesired off-target edits that can confound studies of individual variants. However, in addition to off-target editing, base editing approaches can also introduce bystander changes by modifying unintended bases in an editing window.[48] While base editing systems are limited by the variants that they can generate, hundreds of variants can be made in the native genomic context. For example, recent studies have used base editing to screen over 400 variants in *MYBPC3* and 300 variants in *PARP1*.[49,50] A limitation of base and prime editing is a dependence on unique gRNA binding sites proximal to the desired site of mutation, whereas traditional CRISPR-Cas9 paired with homology-directed repair or serine integrase-based approaches (see below) can modify DNA more distantly.

An alternative to CRISPR-based editing is the use of serine integrases for genomic insertion of transgenes (Figure 2). Serine integrases are site-specific recombinases that mediate the recombination of short (≈50 bp) attB and attP DNA sequences.[51] Plasmids containing an attB sequence can be integrated next to engineered attP sites in the genome, allowing for the insertion of large DNA sequences into a targeted location.[52,53] The attP site can be placed either at the native locus of a gene of interest or in a safe harbor location engineered with

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Table 1. Pooled and Arrayed Functional Assays to Interrogate Cardiac Variants

Assay method	Gene type	For example, genes	Functional alteration	Throughput
MAVE/pooled assays				
VAMP-seq	Any protein-coding gene	*MYH7*[6]	Protein abundance	High
Surface trafficking	Transmembrane proteins	*KCNE1*[7] and *KCNH2*[8]	Protein surface abundance	High
Survival	Protein with an activity that impacts cell viability	*MYH7*,[6] *LMNA*,[9] *SCN5A*,[10] and *KCNE1*[7]	Protein function	High
Splicing	Genes with a loss-of-function mechanism	*SCN5A*,[11] *LMNA*,[12] *MYBPC3*,[12] and *TTN*[13]	Splicing patterns	High
Arrayed assays				
Automated patch clamping	Voltage-gated ion channels	*SCN5A*,[14–17] *KCNQ1*,[16,18,19] *KCNH2*,[20] and *KCNE1*[18]	Ion current amplitude and activation/inactivation properties	Medium-high
Fluorescent plate reader	Ion channels	*KCNH2*,[21] *KCNQ2*,[22] and *TRPM4*[23]	Ion flux changes (eg, Ca^{2+} K^+, and Na^+) via fluorescence indicators	Medium-high
Fractional shortening	Sarcomeric and cytoskeletal genes	*MYH7*,[24] *TNNT2*,[25] *TTN*,[26] and *MYBPC3*[25]	Contraction/relaxation dynamics	Low-medium
Atomic force microscopy	Sarcomeric, cytoskeletal, and nuclear envelope	*LMNA*,[27,28] *DSP*,[29] and *DMPK*[30]	Cell stiffness, contractile force, and viscoelastic properties	Low-medium
Traction force microscopy	Sarcomeric genes	*MYH7*[31]	Force generation against the substrate	Low-medium
Engineered heart tissues	Sarcomeric and cytoskeletal genes	*MYH7*[32–34] and *TNNT2*[84]	Force generation based on post-deflection	Low-medium
Electrical impedance (MEA)	Sarcomeric and signaling genes	*MYH7*,[35] *MYBPC3*,[35] and *TTN*[86]	Contractile kinetics via impedance changes	Low-medium

Low: 1–10; medium: 10–100; and high: >100 variants. MAVE indicates multiplexed assay of variant effect; MEA, microelectrode array; and VAMP-seq, variant abundance by massively parallel sequencing.

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flanking elements that facilitate expression and genetic selection.[54] One commonly used system is a genomic landing pad containing a tetracycline-inducible promoter and an attP site, as well as additional features such as a fluorescent marker or a caspase for genetic selection.[53] AttB-containing plasmids can be integrated into the landing pad, replacing expression of the landing pad genes with a gene of interest and a fluorescent marker from the plasmid. This system enables the identification and selection of on-target integrated versus nonintegrated cells. This system has been used to insert and express thousands of genetic variants in multiplexed analyses.[7,10,53]

Each of the methods described above can generate individual variants for single-variant or arrayed assays or a pool of variants for MAVE experiments. In their multiplexed forms, these methods involve the generation of pools of gRNAs (prime editing and base editing), pools of variant-containing DNA repair templates (saturation genome editing), or pools of variant cDNAs for genomic integration (CRISPR-Cas9 and serine integrases). Pools of bespoke DNAs can be readily synthesized or obtained commercially. Libraries of larger DNAs can also be obtained commercially or can be constructed through polymerase chain reaction using modified primers or by synthesizing short oligonucleotide pools that are subsequently subcloned into the full-length gene.[55,56] Recently developed methods, such as Scalable and Uniform Nicking Mutagenesis and Deep Indel Missense Programmable Library Engineering, promise to increase the efficiency and cost-effectiveness of variant library generation.[57,58]

ARRAYED ASSAYS

Automated Patch Clamping

Genes associated with inherited cardiac arrhythmias primarily encode voltage-gated ion channels.[59] These channels regulate the flux of sodium, potassium, and calcium ions across the cell membrane, generating action potentials that lead to coordinated cardiac contractions. Genetic variants that alter ion flow can disrupt this process, leading to arrhythmias such as long-QT syndrome or Brugada syndrome.[60] By understanding how a variant affects a channel's ability to regulate ion currents, researchers can better predict the pathogenicity of that variant.[61]

Patch clamping is the gold-standard method for characterizing ion channel function and can measure detailed channel properties. For voltage-gated channels, these properties can include peak current and the voltage dependence of channel activation and inactivation.[62] Patch-clamping data of variants with altered functional properties have been widely used to support the pathogenic classification of variants in arrhythmia-related genes.[63] Manual patch clamping is a technique that requires a highly skilled practitioner and is time-consuming.[64] Despite the valuable data provided by manual patch clamping, the time-consuming nature of the procedure has limited its use in high-throughput studies.[64,65]

Automated patch-clamping devices enable the simultaneous testing of many cells.[66] In this planar patch-clamp method, a single cell is suctioned into a

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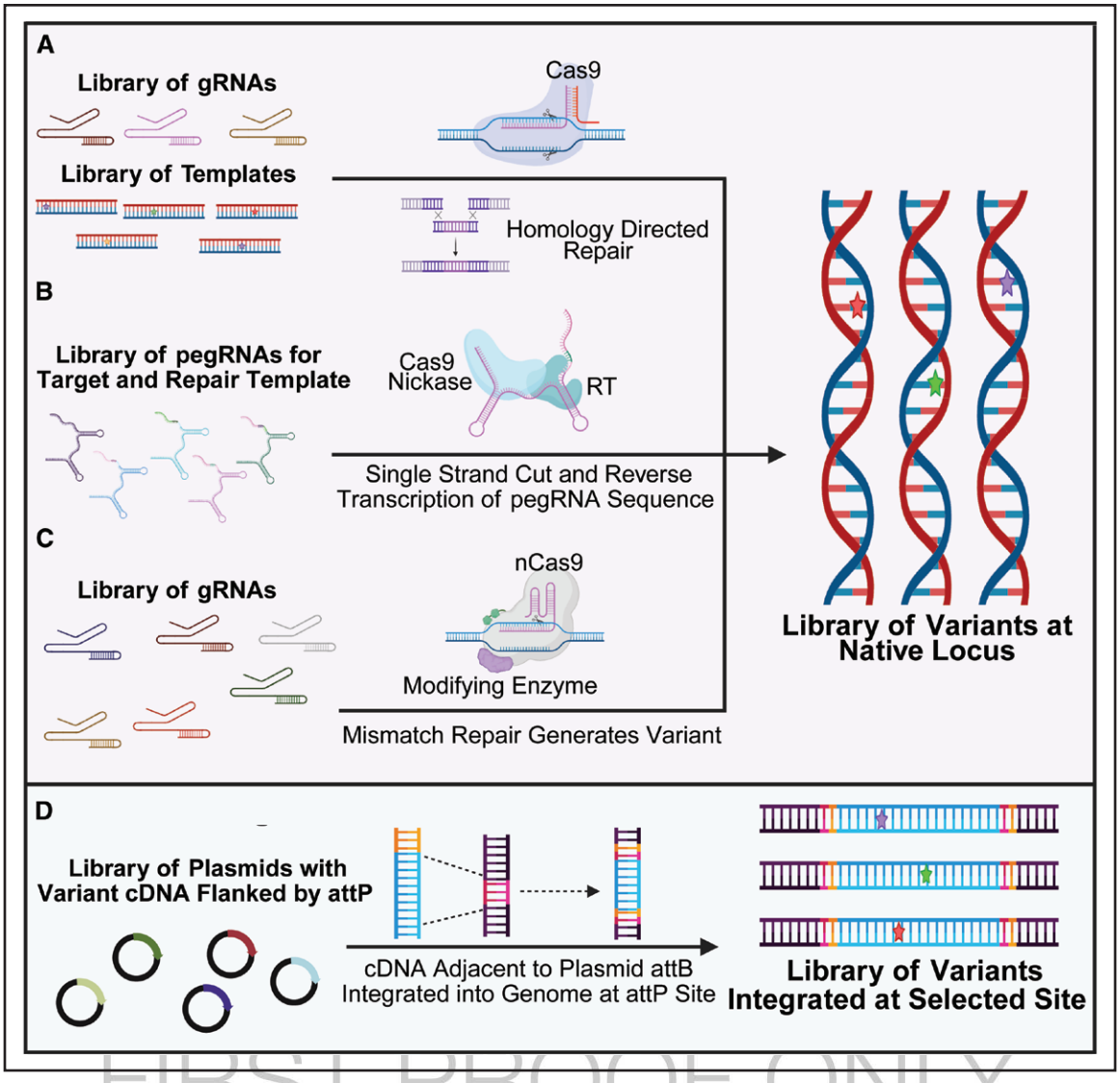

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Figure 2. Commonly used techniques to introduce variants into cells.
A, CRISPR-Cas9 (clustered regularly interspaced short palindromic repeats/clustered regularly interspaced short palindromic repeat–associated 9). Traditional CRISPR systems can use a library of guide RNAs (gRNAs) to target different locations in the genome and a library of templates to generate various variants at each location. **B**, Prime editing systems have a single prime editing gRNA (pegRNA) that determines the location and alteration of the variant. Therefore, a library of pegRNAs can generate a diverse library of variants. **C**, Base editing systems will generate variants based on the modifying enzyme attached. A library of gRNAs can generate a variety of variants in different locations. **D**, Serine integrase recombination. A serine integration system inserts cDNA into a specific part of the genome via recombination. A library of plasmids can, therefore, generate a library of inserted sequences.

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small cavity in each well of a 96- or 384-well plate. The voltage across the cell membrane is manipulated to activate voltage-gated channels, and the resulting currents are measured.[66] This technique can efficiently characterize dozens or hundreds of variants while collecting data of similar quality to that obtained through manual patch clamping.[64] Automated patch clamping has been deployed to characterize over 1300 variants in cardiac ion channel genes, including *KCNQ1*, *KCNH2*, *KCNE1*, and *SCN5A* (Table 2). The high-throughput nature of automated patch clamping is exemplified by 2 recent studies, which tested 533 and 488 *KCNH2* variants.[69,71]

Fluorescent Plate Reader Assays of Ion Channel Variants

High-throughput screening using fluorescent plate readers offers a means of rapidly characterizing the function of ion channel variants.[73] This method can simultaneously record ion channel function across wells in a 96- or 384-well plate. The use of plate readers to measure ion channel activity relies on linking the channel's function to fluorescence changes within each well.[62] This is achieved using ion-sensitive fluorescent indicators.[74] The general method involves (1) distributing cells expressing ion channel variants or drug conditions into different

Table 2. Automated Patch-Clamping Studies of Cardiac Ion Channel Variants

Study	Gene	# Variants	Summary
O'Neill et al (2025)[67]	*SCN5A*	252	Studied all variants found in a large Brugada Syndrome cohort
Vanoye et al (2025)[68]	*KCNE1*	95	Determined population carrier frequency of Jervell and Lange-Nielsen syndrome type 2
O'Neill et al (2024)[69]	*KCNH2*	533	Identified 331 variants with loss-of-function
Ma et al (2024)[15]	*SCN5A*	53	Validated assay with 25 benign and 24 pathogenic controls
Thomson et al (2024)[70]	*KCNH2*	78	Validated assay with 60 controls and evaluated 18 missense variants
Glazer et al (2022)[16]	*SCN5A*, *KCNQ1*, and *KCNH2*	50	Studied 50 variants found from population screening
Ng et al (2022)[71]	*KCNH2*	458	Studied every single nucleotide polymorphism in a hotspot exon
Jiang et al (2022)[72]	*KCNH2*	75	Calibrated assay with 31 controls before testing 44 undefined variants
O'Neill et al (2022)[17]	*SCN5A*	50	Tested dominant negative effects of 35 total and 15 partial loss-of-function variants
Ng et al (2020)[20]	*KCNH2*	30	Evaluated dominant negative effects of 13 pore-domain variants
Kuenze et al (2020)[18]	*KCNQ1* and *KCNE1*	16	Utilized modified cell lines to test dominant negative effects
Glazer et al (2020)[14]	*SCN5A*	83	Validated assay with 30 control variants and then tested 48 VUSs
Vanoye et al (2018)[19]	*KCNQ1*	78	Reclassified 23 or 35 variants of uncertain significance

Studies with at least 10 variants are included. VUS indicates variant of uncertain significance.

wells of a plate, (2) adding ion-sensitive indicators, (3) inducing ion channel activation, and (4) observing fluorescence changes as a proxy for ion concentration.[75] Variants that alter activity compared with the wild type (WT) will cause altered dynamics of cytosolic ion concentrations and altered fluorescence.[75] These methods depend on the use of suitable fluorescent dyes, which are commercially available for all ions relevant to cardiac ion channels. This approach has been deployed for studies of all major ion types (eg, potassium,[22] calcium,[75] and sodium[23] channels), and extensive reviews comparing the different ionic indicators have been written for each ion.[76–78] In certain cases, ions other than the native ion can permeate a channel, such as thallium or rubidium ions passing through open potassium channels.[22] This property is utilized in thallium flux assays, which measure potassium channel activity by detecting thallium ion accumulation with a fluorescent dye.[62,79] A similar technique is applied in rubidium assays, where the passage of rubidium ions through potassium channels is quantified using spectrophotometry.[22]

RYR2 encodes a sarcoplasmic reticulum–localized calcium channel that plays a critical role in excitation-contraction coupling. Gain-of-function variants in *RYR2* are associated with catecholaminergic polymorphic ventricular tachycardia, whereas loss-of-function variants can cause the recently described calcium release deficiency syndrome.[80] *RYR2* function has been studied using fluorescence resonance energy transfer assays to analyze its interactions with calmodulin and FKBP12.6.[81] In addition, *RYR2* activity has been investigated using R-CEPIA1er, a calcium sensor localized to the endoplasmic reticulum.[82] To date, these approaches have been used mainly for drug screening in plate reader formats. However, because differentiating between loss-of-function and gain-of-function *RYR2* variants is critical for understanding *RYR2*-related arrhythmias, we anticipate the growing use of high-throughput approaches to study *RYR2* variants.

Fluorescent plate reader measurements of ion channel function are more widely available than automated patch-clamping machines. However, a limitation of this method is that it typically does not determine precise ion channel properties such as activation or inactivation voltages.[62] While most high-throughput studies of this type are utilized for drug screening, these methods can be easily adapted for testing the effects of different variants; instead of each well containing a different channel modulator, each well can include a different variant. In summary, fluorescent plate reader assays offer a versatile and accessible method for assessing ion channel function, with applications ranging from drug screening to variant effect testing.

CARDIOMYOCYTE MECHANICS

Cardiomyocyte contractile dysfunction is closely associated with cardiac disease. Mechanical properties of cardiomyocytes altered in cardiomyopathy include active and passive force, contraction and relaxation time, and cell adhesion.[83] In this section, we describe a series of methods to assess contractile function, which can aid in characterizing variant pathogenicity and phenotype.[84] These methods can be adapted for single cells, cardiomyocyte monolayers, cardiac embryoid bodies, and engineered heart tissues (EHTs), with varying throughputs and applications (Table 3; Figure 3). Typically, single-cell and monolayer methods offer higher throughput

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Table 3. Functional Mechanical Readouts

Technique	Throughput	Single cells	Monolayers	Tissues	# Variants reported	References
AFM	High	*	*		3	28–30
TFM	High	*	*		2	31,85,86
Sarcomere fractional shortening	High	*	*		3	87
Microposts	Medium	*			2	88,89
Electrical impedance	Medium		*		6	35
Isolated myofibrils	Low	*			2	90
EHTs	Low			*	5	91

AFM indicates atomic force microscopy; EHT, engineered heart tissue; and TFM, traction force microscopy.

compared with tissue-based methods due to the larger cell numbers and hands-on time required to generate EHTs. However, recent advances in tissue miniaturization have increased the throughput of tissue mechanics testing, enabling their use for variant effect assessments.[92]

Fractional Shortening

As a cardiomyocyte contracts, both cell length and sarcomere length cyclically decrease. Fractional shortening measures the change in cell or sarcomere length normalized to the resting length. Both cell and sarcomere length can be detected with brightfield microscopy, enabling simple measurement across hundreds of cells.[93] Sarcomeres can also be tracked by incorporating fluorescent proteins using gene editing, small molecules, or viral transduction.[60,94] Cell membranes and sarcomeres can be tracked across video frames using tools such as SotaTool,[95] SarcTrack,[96] and SarcGraph,[97] or in proprietary software from IonOptix.[98] Notably, fractional shortening is an indirect measure of contractile force and can be a poor predictor of force in certain disease contexts.[87]

Atomic Force Microscopy

Atomic force microscopy (AFM) is a microscopy technique that uses a nanometer-scale metal tip attached to a flexible cantilever. When the tip makes contact with a cardiomyocyte, the cantilever is deflected by a certain distance. This deflection corresponds to piconewton-scale force measurements, determined by the cantilever's spring constant. AFM can, thus, capture the mechanical properties of cardiomyocytes, such as cell stiffness and stress relaxation. Beat rate, contractile force, and contractile kinetics can be measured by locking the tip at the cell surface and measuring its deflection as the cell contracts. AFM was first applied to rat atrial myocytes in 1995[99] and to human induced pluripotent stem cell–derived cardiomyocytes (hiPSC-CMs) in 2012.[100] AFM is well-suited for studying contractile dynamics of single cells with high resolution.[90,101,102] The technique has been used to investigate the effects of variants in cardiomyopathy-related genes such as *LMNA*, *DSP*, and *DMPK* on cardiomyocyte mechanics.[28–30] Approximately 100 force tip indentations can be measured per hour, enabling rapid assessment of single variants in an arrayed format.[103]

Traction Force Microscopy

Traction force microscopy is a fluorescence microscopy technique used to measure the contractile mechanics of cardiomyocytes. In traction force microscopy, fluorescent spheres ranging from 100 to 1000 nm in diameter are embedded in a polymer hydrogel with adjustable stiffness. An attachment protein matrix is then patterned on the hydrogel using polydimethylsiloxane stamps or photolithography. Depending on the pattern, cells can attach to the coated surface as single cells, a monolayer, or a tissue.[104,105] When cardiomyocytes contract, they exert force on the underlying hydrogel, causing displacement of the embedded fluorescent beads. By capturing videos of these displacements, researchers can quantify traction force and contractile kinetics as a function of the hydrogel's Young modulus. This approach has been successfully used with hiPSC-CMs expressing HCM-causing *MYH7* variants p.Arg403Gln, p.Pro710Arg, and p.Met852Thr, demonstrating that these variants are hypercontractile.[31,85,86]

Microposts

In the micropost technique, cardiomyocytes are cultured on arrays of elastomeric posts coated with extracellular matrix.[88,89,106–108] These posts deflect under cell tension, which can be converted to force measurement based on the physics of the bending posts. This principle has been adapted to measure the mechanical properties of cardiomyocytes.[88,89] However, a significant limitation is the inability to electrically stimulate the cardiomyocytes when they are cultured on the microposts.[88]

Engineered Heart Tissues

In vitro tissue approaches attempt to mimic the native cardiac architecture to capture macroscopic properties

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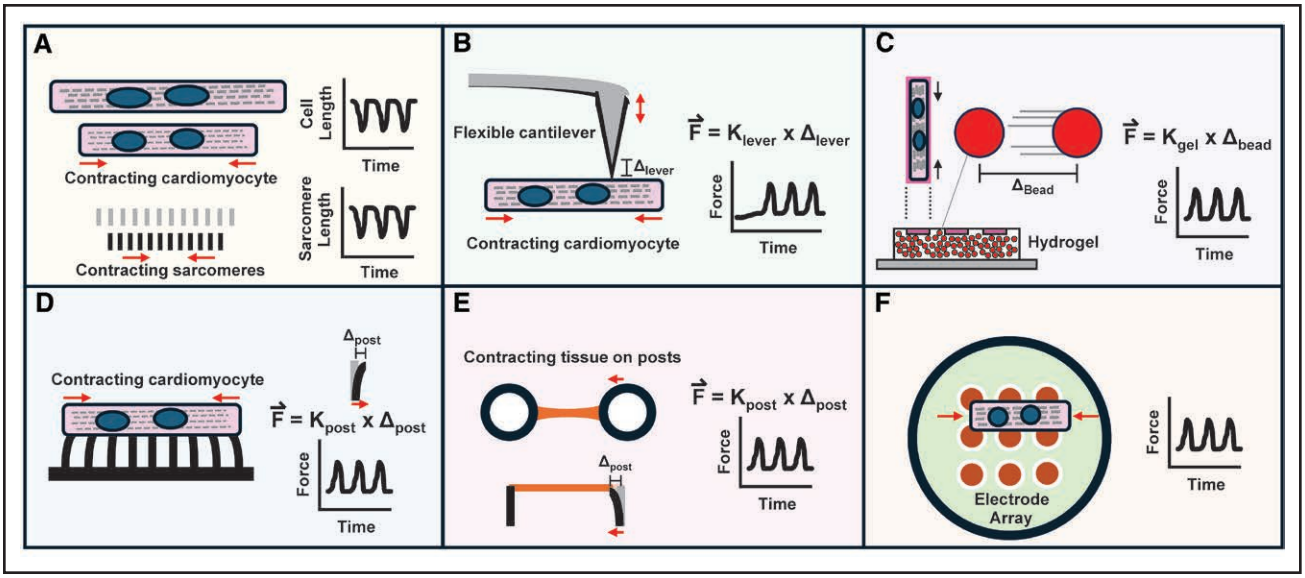

Figure 3. Methods for mechanical assessment of cardiomyocytes.
A, Fractional shortening measures the relative change in cell length or sarcomere length in contracting cardiomyocytes. **B**, Atomic force microscopy uses a flexible cantilever in contact with a contracting cardiomyocyte to estimate local contractile force based on the deflection of the cantilever and the cantilever's stiffness. **C**, Traction force microscopy uses a hydrogel with embedded fluorescent beads. As cardiomyocytes contract, force can be estimated based on the displacement of beads beneath the cardiomyocyte as a function of hydrogel stiffness. **D**, Cardiomyocytes can be seeded on flexible microposts of defined stiffness, enabling the measurement of force based on post-deflection. **E**, Engineered heart tissues use a 3-dimensional tissue seeded across one flexible and one rigid post. Tissue contraction deflects the flexible post, permitting force measurement. **F**, In impedance assays, cardiomyocytes are seeded on top of microelectrode arrays. As cardiomyocytes contract, the electrical resistance decreases, a proxy for contractile function.

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that are absent in single-cell or monolayer contexts. The most established method in this category is the EHT.[91] The basic principle involves coculturing cardiomyocytes with stromal cells in a slurry, which is then seeded onto hydrogels or scaffolds, such as flexible posts, to form tissue constructs. As the tissues contract, the attached posts deform, enabling measurement of contractile force. Contractile function of large EHTs can also be assessed by directly connecting the tissues to a force transducer. Early strategies using EHTs relied on tissues comprised of 10^5 to 10^6 cardiomyocytes per tissue, significantly limiting the scale of throughput across variants.[109,110] Recently, miniaturized or micro-EHTs have been developed, which mimic the contractile properties of larger EHTs while requiring as few as 16 000 cardiomyocytes per tissue.[92,111] These micro-EHTs offer scalability across variants, as they can be cast into individual wells of a 96-well plate, enabling parallel assessment of many tissues at once. The Curi Bio Mantarray system uses EHTs attached to posts with embedded magnets for label-free, electromagnetic measurement of contractile kinetics in response to electrical stimulation, enabling simultaneous measurement of up to 24 EHTs.[112]

Electrical Impedance With Microelectrode Array

Label-free electrical impedance is a well-established method for sensing cell behavior. Microelectrode arrays embedded in tissue culture plates can detect changes in impedance and correlate these changes to cardiomyocyte mechanical output, as described by a power law relationship.[113] Impedance measurements have been used with hiPSC-CMs to assess cardiotoxicity in response to putative therapeutics.[114] Furthermore, impedance measurements have recently been used to assess the efficacy of myosin ATPase inhibitors in restoring normal contractile function in hiPSC-CMs with HCM-causing *MYH7*, *MYBPC3*, and *TNNI3* variants.[35] This approach shows promise for high-throughput mechanical assessment of variant-expressing cells in a 96-well plate format.

POOLED ASSAYS

In contrast to arrayed assays where each variant is studied individually, MAVEs involve the pooled creation and joint characterization of a library of variants. The general approach involves (1) saturation mutagenesis of a target gene or region, (2) expression of pools of mutants in an appropriate cell model, (3) en masse multiplexed selection for cellular phenotypes that reflect variant effects on relevant protein function(s), (4) next-generation sequencing to compare variant abundance in preselection and postselection pools, and (5) data analysis and display. The most critical step of an MAVE is to develop an assay that faithfully models the disease process of interest. Several assay types are described in the following.

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Flow Cytometry–Based Assays

Flow cytometry combined with cell sorting enables the rapid and high-throughput separation of cells into bins spanning a broad range of phenotypes. Each bin, characterized by a distinct phenotypic range, undergoes deep sequencing to quantify the frequency of each variant within that bin. By analyzing variant frequencies across all bins, researchers can reconstruct an estimated phenotypic profile for each variant, allowing for the detection of variant profiles that significantly diverge from controls. Flow cytometry–based assays for multiplexed functional assays rely on measuring single-cell phenotypes. These include cell size, expression of a fluorescent reporter, or an epitope prelabeled with fluorophore-conjugated antibodies. Figure 4 illustrates a general workflow for 3 different fluorescence-based flow cytometry MAVEs. Examples include (1) variant abundance by massively parallel sequencing, where the intensity of GFP (green fluorescent protein) fused to a protein of interest serves as a readout of protein expression,[115] (2) trafficking assays using fluorescent antibody staining to assess surface protein abundance,[116] and (3) assays measuring protein function that use fluorescent markers, where fluorescence changes occur as a downstream response to activity of the protein of interest.[117]

Arguably, the most fundamental way to assess the effect of a genetic variant is to determine whether it alters the abundance of its associated protein product. Pathogenic variants in many genes can reduce protein abundance by destabilizing the mRNA or protein produced, triggering nonsense-mediated decay of the mRNA, or through other mechanisms.[118] Matreyek et al[115] utilized this principle to develop a flow cytometry–based multiplexed assay technique called variant abundance by massively parallel sequencing. This method can detect deleterious variants that alter protein abundance.

Variant abundance by massively parallel sequencing involves tagging the protein of interest with a fluorescent marker, allowing a cell's fluorescence intensity, measured via flow cytometry, to serve as an indicator of the protein's abundance. A pooled library of tagged protein variants can be binned by fluorescence intensity using flow sorting, and then, each bin can be sequenced to quantify the variant's impact on protein abundance.

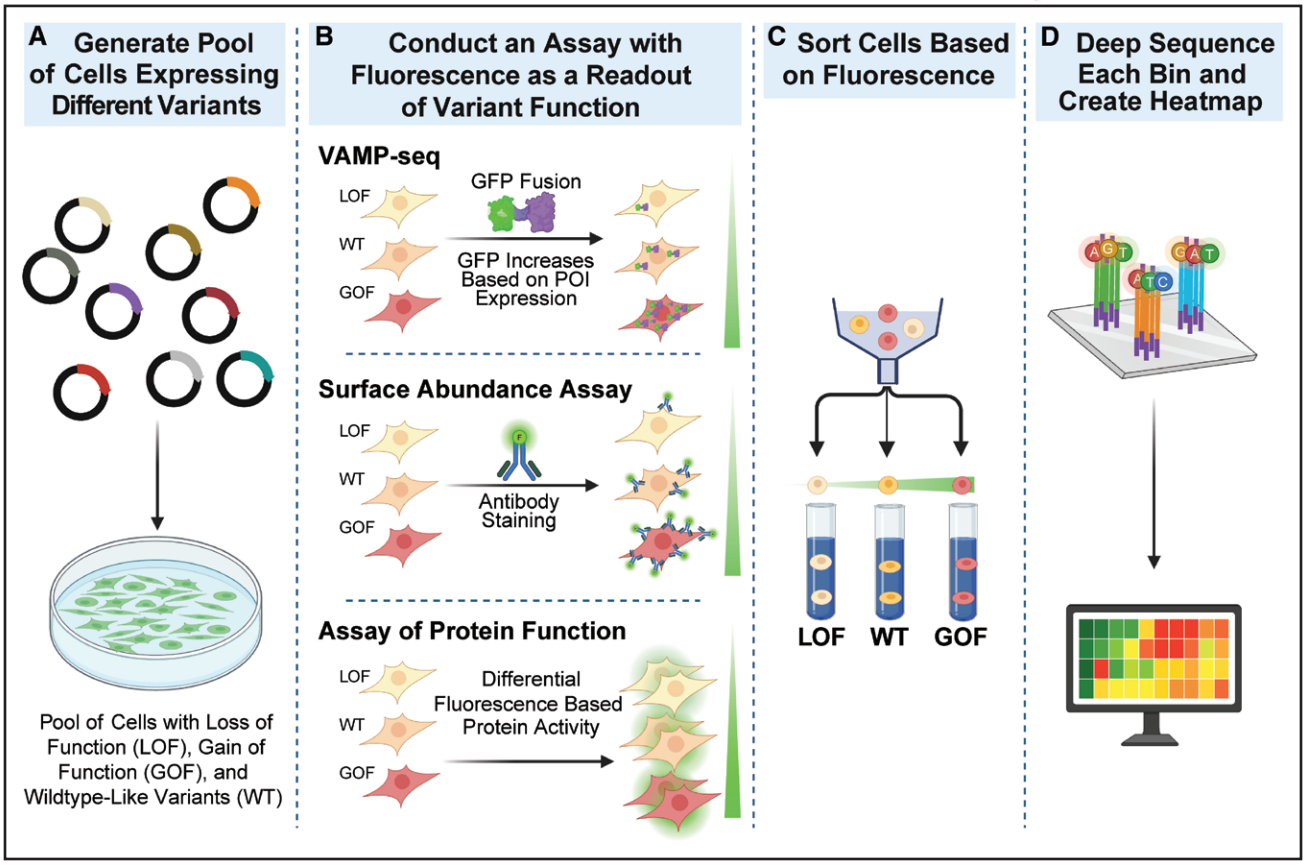

Figure 4. Fluorescence-based multiplexed assays of variant effect.
A, Generate a pool of cells expressing different genetic variants. **B**, An assay is performed where cellular fluorescence is linked to protein function. **C**, Flow sorting is used to separate cells based on their fluorescence intensity. **D**, Each bin is deep sequenced to generate functional scores for each variant, which are presented as a heatmap. GFP indicates green fluorescent protein; and VAMP-seq, variant abundance by massively parallel sequencing.

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This method enables high-throughput assessment of protein levels using flow cytometry for fluorescence-based cell sorting, followed by next-generation sequencing and abundance calculations. Friedman et al[6] applied variant abundance by massively parallel sequencing to evaluate a library of 113 heterozygous *MYH7* variants in hiPSC-CMs, by measuring the abundance of β-MHC (β-myosin heavy chain), the protein encoded by *MYH7*. This approach facilitated the identification of pathogenic variants associated with low β-MHC abundance. Similarly, Yamamoto et al[49] used direct antibody-labeling of a library of *MYBPC3* variants in hiPSC-CMs, followed by flow sorting and sequencing, to identify *MYBPC3* variants disrupting protein abundance.

Additional CVD phenotypes, such as enlarged cardiomyocyte size or increased expression of the heart failure marker BNP, can be detected using flow cytometry–based assays. For instance, Pettinato et al[119] used CRISPR/Cas9 to knock in a tdTomato reporter into the endogenous *NPPB* locus of a *TNNT2* knockout hiPSC line. They then transduced hiPSC-CMs derived from this line with 51 distinct *TNNT2* variant lentiviruses and used flow cytometry to measure tdTomato as an indicator of *NPPB* expression. *TNNT2* variants known to cause HCM exhibited significantly increased *NPPB* expression, whereas variants causing dilated cardiomyopathy showed decreased expression. In the recent MYBPC3 MAVE study, the authors also used a BNP reporter to assess cardiomyocyte hypertrophic signaling across the pool of variants.[49]

To enhance flow cytometry–based functional assessments of CVD-associated variants, additional pathogenic markers that can be fluorescently labeled, either through immunostaining or genetically encoded reporters, should be identified. These markers would enable evaluations in an arrayed format or, if the marker's dynamic range is sufficiently broad, in a pooled format. Alternatively, simple morphological parameters such as cell size could be assessed by flow cytometry, eliminating the need for fluorescent labeling. Indeed, several groups have used flow cytometric measurement of forward scatter as a proxy for cell area to evaluate variants causing HCM and dilated cardiomyopathy.[33,119] However, forward scatter assessments have only revealed minor (<10%) differences in relative cell size between pathogenic and WT controls. This observation is corroborated by absolute, flow cytometry–based volumetric assessments of hiPSC-CMs, which show a modest increase in cell volume (12%–51%).[110] Collectively, these findings suggest that enhancing hypertrophy phenotypes through various growth condition modifications would be necessary to expand the dynamic range for pooled forward scatter assays.

Trafficking Assays

Of the cardiac ion channels, *KCNH2* variants have been the most extensively studied using solubility and glycosylation assays, which serve as proxies for evaluating protein abundance and surface expression.[120,121] These studies established that most loss-of-function *KCNH2* variants impair protein stability and localization. They further led to a general appreciation that variants in transmembrane proteins that disrupt overall protein folding, stability, and trafficking to the membrane are usually the major mechanism of loss-of-function.[122] Building on this work, MAVE-compatible fluorescent assays of surface abundance were developed (Figure 4).[8] This approach uses fluorophore-conjugated antibodies targeting extracellular epitopes (native or epitope tagged) to label only proteins at the cell surface, leaving intracellular pools unlabeled. Each variant's surface abundance is quantified by fluorescence-activated cell sorter-sorting cells into fluorescence bins and sequencing the barcodes or variants from each bin.[123] Surface abundance scores are assigned to each variant based on its frequency in low-, WT-like, or high-fluorescence bins. This assay has been deployed to generate surface abundance scores for thousands of variants across cardiac ion channel genes, *KCNE1*,[7] *KCNJ2*,[124] and *KCNH2*.[69] Notably, O'Neill et al[69] performed an MAVE analysis of surface expression of over 17 000 *KCNH2* variants; this data set was also correlated with a large automated patch-clamp data set and clinical arrhythmia outcomes. As an alternative method for measuring cell surface expression, researchers assessed ion channel trafficking using a pH-sensitive fluorescent tag, pHluorin, which distinguishes between internal and plasma membrane–localized channels based on fluorescence changes at different pH levels. Lowering the extracellular pH from 7.4 to 6.0 caused a significant reduction in fluorescence for variant proteins located at the cell surface, while internal variant proteins remained unaffected.[125] We note that it is possible for a variant to be correctly trafficked to the surface yet still hinder function. Although a surface abundance assay alone may not identify this class of variants, pairing this assay with a functional assay can allow researchers to gain insights into the mechanisms underlying a variant's pathogenicity.[7]

Key considerations for this assay are that each cell should only express a single variant and that the overall expression of the ion channel must be relatively consistent. This can be achieved using specialized cell lines that allow for controlled cDNA integration or appropriate CRISPR systems.[43,53] Alternatively, coexpressing a fluorescent protein with the ion channel gene can provide a means to normalize trafficking scores to transgene expression.

Survival Assays

Survival assays that compare variant frequencies within a pooled variant library before and after a selection pressure are perhaps the most accessible high-throughput

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phenotyping strategy. Many cardiac diseases, such as dilated cardiomyopathy, inherited arrhythmia syndromes, and HCM, feature reduced cardiomyocyte survival relative to WT cardiomyocytes.[126] Pathogenic variants can exert damaging effects through mechanisms beyond cell death; however, cells that cannot survive cannot function.

Pathogenic variants in numerous CVD-related genes have been shown to cause reduced cardiomyocyte survival, priming these genes for survival-based MAVEs. For example, heterozygous premature truncating variants in the *DSP*, which encodes an intercalated disk protein, are known to cause a subtype of arrhythmogenic cardiomyopathy referred to as desmoplakin cardiomyopathy, often characterized by cell death.[127] Indeed, cardiomyocyte-specific deletion of *Dsp* in mice causes a robust PANoptosis phenotype (apoptosis, necroptosis, and pyroptosis).[127,128] Interestingly, the location of premature truncating variants in the *DSP* gene plays a role in the development of ventricular arrhythmias,[129] suggesting that other *DSP* phenotypes, such as reduced cell survival, could also be domain-dependent and benefit from high-throughput survival-based assessments in the future. In addition, heterozygous missense variants in the nuclear inner membrane gene *LMNA* cause a highly penetrant dilated cardiomyopathy subtype, with associated arrhythmias. This syndrome is often referred to as laminopathy and also features cell death.[130] For example, cardiomyocyte-specific expression of the p.Asp300Asn pathogenic variant of *LMNA* in mice causes a robust increase in TUNEL (terminal deoxynucleotidyl transferase dUTP nick end labeling)-positive cells and upregulation of *Tp53* signaling cascades,[130] suggesting that a high-throughput survival-based assay could help to map variant effect across *LMNA*. A comprehensive study of 178 *LMNA* missense variants utilized aggregation and solubility assays, with further validation in iPSC-CMs. Notably, the study revealed that pathogenic variants often lead to lamin A aggregation in the nucleus, indicating that these diseases are predominantly driven by protein misfolding.[9]

Pathogenic missense variants in the sarcomeric thick filament gene, *MYH7*, are a leading cause of HCM. Previous reports have shown that single pathogenic *MYH7* variants (p.Arg723Cys and p.Glu848Gly) can cause reduced survival in hiPSC-CMs.[33,131] Friedman et al[6] leveraged this observation to establish a multiplexed cardiomyocyte survival assay to functionally assess heterozygous *MYH7* missense variants in hiPSC-CMs. This assay involved performing next-generation sequencing of the *MYH7* variant library in hiPSCs and hiPSC-CMs. Frequencies of each variant were then calculated and compared between hiPSC-CMs and hiPSCs to discover which variants became prevalent after 47 days of culture in vitro. This analysis revealed that all 3 pathogenic variants, p.Glu848Gly, p.Met852Thr, and p.Met852Lys, had lower survival than 3 synonymous variants, p.Ser866=, p.Ala850=, and p.Glu848=(assumed to be functionally normal); a similar analysis of day 25 hiPSC-CMs revealed a similar outcome.[6] In addition, 2 pathogenic variants were lost from the day 47 library hiPSC-CMs (p.Lys865Arg and p.Met852Arg), indicating more severe cases of this survival phenotype.[6] This study serves as proof of concept that multiplexed survival assays can be leveraged for accurate functional annotation of additional *MYH7* (and other genes) variants in hiPSC-CMs in the future.

Survival assays have been deployed in ion channel MAVEs to obtain functional scores in HEK293T cells for a 36-nucleotide region in *SCN5A* and nearly every variant in *KCNE1*.[7,10] Glazer et al[10] modified a previously developed assay that exposes cells to the sodium channel agonists, veratridine and brevetoxin, and the Na^+/K^+ exchanger inhibitor ouabain, to assess the function of a library of *SCN5A* variants; loss-of-function *SCN5A* variants resisted sodium influx and sensitivity to sodium ion overload, leading to increased (or unchanged) survival, whereas WT-like *SCN5A* variants showed reduced viability relative to pretreatment. A similar loss-of-function assay developed by Muhammad et al[7] tested the hypothesis that long-term (8-day) expression of WT *KCNE1* (in the context of a *KCNQ1* gain-of-function variant p.Ser140Gly) would decrease HEK293T viability relative to preexposure due to increased potassium ion efflux. Application of this assay to a library of *KCNE1* variants revealed 401 partial- and 173 loss-of-function variants with unchanged survival.[7] Other ion channel genes could likely be studied with similar strategies.

Splicing Assays

Up to 10% of pathogenic variants in Mendelian diseases disrupt RNA splicing, highlighting the importance of studying this type of variant.[132] However, assays that measure variant effects often use plasmids or DNA sequences containing only the coding sequences of a gene, which do not undergo RNA splicing, thereby preventing the identification of variants that alter RNA splicing. One approach to studying splice-disrupting variants is the use of minigene assays, in which DNA containing an exon or exons of interest, along with surrounding intronic sequences, is transfected into cells.[133] Normal or abnormal splicing can then be measured by RNA sequencing or reverse transcription polymerase chain reaction and gel electrophoresis.[12,134,135]

Recent efforts have scaled up this type of minigene assay to allow for multiplexed studies of pooled variants. Variant exon sequencing is a method in which a barcoded minigene system containing a library of variants is transfected into cells and given time for RNA processing to occur.[136] RNA is then extracted from the cells, reverse-transcribed into cDNA, and subjected to deep sequencing to determine the impact of each variant on splicing. In the original study, this system measured 2059 variants spanning 110 exons.[136] ParSE-seq is a similar

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process that utilizes a minigene system with barcoded RNA to measure variant effects via RNA sequencing after transfection.[134] This method was used to study 224 *SCN5A* variants in hiPSC-CMs. A similar pooled minigene method was used to measure 196 candidate splice-altering variants in *LMNA* and *MYBPC3*, which are associated with dilated cardiomyopathy and HCM, as well as *TTN* truncating variants that may contribute to dilated cardiomyopathy.[12,13] The scalability of these approaches has enabled the comprehensive testing of all possible splicing variants in targeted exons of cardiac genes, such as *POU1F1*, *MYBPC3*, and *FAS*.[49,137,138] An alternate minigene system relies on the introns and exon of interest being flanked by incomplete sequences of GFP. If the exon is skipped during splicing, cells will transcribe and translate GFP, allowing them to fluoresce. Fluorescent cells can then be isolated via flow sorting and subjected to deep sequencing. This method was used to measure splicing in more than 25 000 pooled variants.[139]

Evidence Strength Calculations

If functional assays are to be used for clinical variant classification, current ACMG guidelines recommend that assays be calibrated using a large set of known benign and pathogenic variants.[140] The assay's evidence strength is determined by the number of control variants and the assay's ability to distinguish benign from pathogenic variants. A Bayesian calculation of the odds of pathogenicity (termed OddsPath) can then be converted into ACMG evidence strengths (eg, supporting, moderate, or strong) for variant classification.[140,141] To improve OddsPath estimates for functional assays, we recommend incorporating as many benign and pathogenic controls as is feasible. A key advantage of MAVEs is that they generate functional data for large numbers of control variants, enabling the possibility of stronger OddsPath evidence codes.

Assays of the Future

This review has highlighted current methods for assessing variant effects in large classes of cardiac disorders, including inherited arrhythmia syndromes and HCM. We now outline future directions for high-throughput assays to enhance our understanding of genetic variation in CVDs.

Broader Contexts for Arrhythmia Assays

As outlined above, arrhythmia-causing variants in voltage-gated ion channels have been effectively studied using automated patch-clamping and MAVE experiments. However, much of this work has relied on heterologous expression systems, such as HEK293 cells with exogenous expression of a single cDNA isoform of each gene. While these approaches have been highly successful in identifying channel-altering variants, an important question remains: to what extent do accurate assessments of variant effects require a disease-relevant cellular context and endogenous gene expression? Future studies focusing on arrhythmia-related genes in more native contexts, such as gene-edited iPSC-CMs, and follow-up investigations of variants with discordant results between functional data and patient phenotypes, could help address this gap. Multiple gene-editing approaches in iPSC-CMs have already been applied to study dozens to hundreds of variants in genes such as *MYH7*,[6,86] *MYBPC3*,[49] and even *KCNH2*,[54] demonstrating the potential to expand functional models for arrhythmia-associated genes. In addition, studies in heterologous systems could be extended to a broader set of variant contexts, incorporating features such as coexpression of additional binding partners, heterozygous expression, common polymorphisms, splice isoforms, or response to drug exposures.

Scaling Up Mechanical Assays

Conventional wisdom holds that cardiomyopathy genes, which intricately alter contractile force dynamics, are best studied in cardiomyocytes with functional sarcomeres. As we outlined above, a broad set of mechanical assays can be deployed to assess variant effects and elucidate disease mechanisms. However, most studies utilizing these assays have been conducted in arrayed formats, which are often limited in throughput. Therefore, a critical future direction is to increase the throughput of arrayed assays and to develop innovative, MAVE-compatible assays for cardiomyopathy genes that capture key disease processes. In addition, efforts are underway to enhance the efficiency and scalability of gene editing in iPSC-derived cardiomyocytes. Finally, it may be possible to investigate some functions of cardiomyopathy-related genes in non-cardiomyocyte contexts, enabling the use of simpler heterologous cellular models.

PERSPECTIVES AND CONCLUSIONS

High-throughput methods for functional assessment have transformed the study of genetic variants implicated in cardiac diseases, enabling unprecedented insights into variant-disease relationships and the molecular mechanisms of conditions such as inherited arrhythmias and cardiomyopathies. While scalable techniques such as MAVEs and automated patch clamping have addressed critical challenges such as the classification of VUS, the field remains constrained by the need for disease-relevant models such as hiPSC-derived cardiomyocytes and pooled assays for physiologically relevant phenotypes. Expanding the range of phenotypes studied and linking these functional insights to clinically actionable

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outcomes will be essential for improving diagnostic accuracy and therapeutic decision-making. Despite these challenges, continued innovation promises to further advance variant classification, refine precision medicine approaches, and ultimately improve patient care in cardiovascular genetics.

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Disclosures

None.

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